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Using Smart Glasses for Monitoring Cyber Threat Intelligence Feeds

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***Abstract*—The surge of COVID-19 has introduced a new threat surface as malevolent actors are trying to benefit from the pandemic. Because of this, new information sources and visualization tools about COVID-19 have been introduced into the workflow of frontline practitioners. As a result, analysts are increasingly required to shift their focus between different visual displays to monitor pandemic related data, security threats, and incidents. Augmented reality (AR) smart glasses can overlay digital data to the physical environment in a comprehensible manner. However, the real-life use situations are often complex and require fast knowledge acquisition from multiple sources. In this study we report results from an experiment with six subjects using an AR overlaid information interface coupled with traditional computer monitors. Our goal was to evaluate a multi tasking setup with traditional monitors and an AR headset where notifications from the new COVID-19 MISP instance were visualized. Our results indicate that better situational awareness does translate to increased task performance, but at the cost of a gender gap that requires further attention.**

***Index Terms*—Covid-19, Smart Glasses, HoloLens 2, MISP**

I. Introduction

The outbreak of COVID-19 has introduced mandatory lockdown policies and strict social distancing protocols that have majorly interrupted social economic life in most countries. Along with the health crisis, we have also seen a surging information crisis where large volumes of information across multiple domains urgently need to be shared, analysed, authenticated, and verified. In this backdrop, many pre-existing cybersecurity challenges have been exacerbated and often morphed into complex hybrid phenomena. In other words, many of the threats we are now facing are not new, but new tools are needed in fighting those disruptions [4], [15].

Past research has shown that smart glasses have an impact on comprehension [6]. Augmented reality (AR) allows embedding semantically familiar material such as objects to an environment while retaining the spatial information. This can enhance learning experiences as it complements perception. However, the real benefit of AR and any cross reality lies in the possibility to visualize abstract digital data with spatial information, to aid in both comprehension and situational awareness [1], [10]. While the aerospace and defense sectors are utilizing the technology, the uptake of AR by the cybersecurity community is still lagging. Given the lack of research in this area, the current experiment was driven by the following research question: Does the consolidation of cyber threat intelligence feeds into a single display improve information comprehension?

II. Applications and Tools

A. TIPs and the MISP

With the increased usage of internet, the type and number of cyber attacks has also increased. This spawns from the fact that the main purpose of internet originally was connectivity and not security. Over time, the nature of these cyber threats has changed in terms of scope, impact, and complexity. This change has also created an urgent need to develop information sharing platforms to share threats and mitigation strategies with each other, in order to help protect porous infrastructures against attacks. In response to this, new tools have been introduced by the software industry known as Threat Intelligence Platforms (TIP), enabling organizations to share threat information with the community and help mitigate risks. Threat intelligence in simple terms can be described as the knowledge and information of threats, its capabilities, infrastructure, motives, goals, and resources. Threat Intelligence Platforms are the tools that help in getting the threat

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intelligence information into the form of security data, and then analyse it to identify and mitigate security threats [11]. TIPs can be classified into two major categories based on their availability as Open source (free) or commercial (paid). Alien Vault open threat intelligence (OTX) and Malware information Sharing Platform (MISP) are free to use, while commercial TIPs include the Accenture cyber threat intelligence platform, Anomali ThreatStream, IBM-Xforce Exchange etc. These platforms have different agendas and goals such as providing threat sharing, sharing aggregated data, indicators of compromise, and supporting different standards and licensing models [18].
Currently the most highlighted and widely used threat intelligence platform is the Malware Information Sharing Platform (MISP). It is a community and government-driven open source trusted platform developed by The Computer Incident Response Center Luxembourg (CIRCL). In MISP, information usually refers to Indicators of Compromise (IOC) and information about cybersecurity threats and incidents. Furthermore, MISP provides information about vulnerabilities and financial indicators used in fraud cases. An entry in MISP is known as an event, containing the information, IOCs, and attachments in the form of attributes (category and type). MISP has several sharing models such as sharing information with organization only, community only, with some connected communities, or with all MISP communities. MISP has its own taxonomy for sharing information and tagging them correctly. It has a REST client and an API for data pulling and pushing purposes. MISP has many instances, events, and attributes and is widely used for data extraction and research. [21].

B. The COVID-19 MISP instance

Since December 2019, the world has been struggling with a COVID-19 pandemic. The effect of COVID-19 on the entire world has been immense and it has also affected every sector of the world's economy. Many organizations and researchers took advantage of this time and tried to explore and track COVID-19 data in real time by visualizing them using dashboards. These dashboards make it easy for the layman to see and understand the day-to-day numbers. But over time, cyber criminals also moved into the world of COVID-19 and began launching attacks, scams, spreading misinformation and fake data. To mitigate any risks, CIRCL introduced a new retrofitted instance of MISP that contains feeds and events related to COVID-19 [13]. The instance also includes a custom dashboard with official statistics for COVID-19 deaths, confirmed cases, and recovered cases by country (Fig. 1). The events home page is the same as the MISP main instance but with COVID-19 based event content. On this page users can find all the events with fake/abusive content related to COVID-19. These events include misinformation, fake data, fake COVID-19 domains and breaching alerts. It is also possible to extract data from this instance either by using a cURL command or by using a Python library called PyMISP that enables access to MISP platforms via their REST API. For both cURL and PyMISP, it is mandatory to mention the MISP instance URL parameters and the unique key of the user to download the data either in JSON or comma separated values (CSV) format [20]. The extracted data includes events and their attributes (Fig. 2). The attributes include a unique id, event information, analysis, threat level id etc.

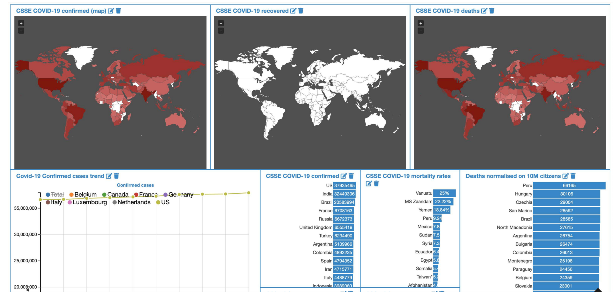
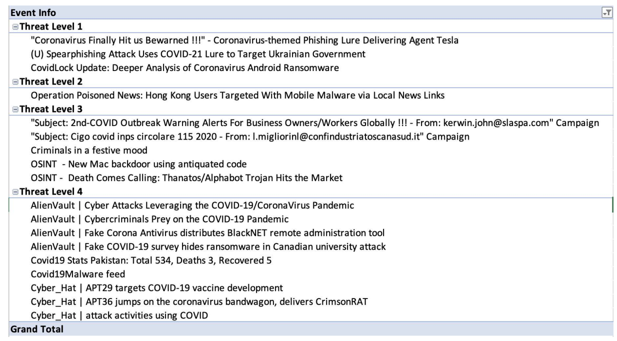


Fig. 1. MISP Covid-19 Dashboard



Event Info
Threat Level 1
"Coronavirus Finally Hit us Bewarned !!!" - Coronavirus-themed Phishing Lure Delivering Agent Tesla
(U) Spearphishing Attack Uses COVID-21 Lure to Target Ukrainian Government
CovidLock Update: Deeper Analysis of Coronavirus Android Ransomware
Threat Level 2
Operation Poisoned News: Hong Kong Users Targeted With Mobile Malware via Local News Links
Threat Level 3
"Subject: 2nd-COVID Outbreak Warning Alerts For Business Owners/Workers Globally !!! - From: kerwin.john@slaspa.com" Campaign
"Subject: Cigo covid inps circolare 115 2020 - From: l.migliorinl@confindustriatoscanasud.it" Campaign
Criminals in a festive mood
OSINT - New Mac backdoor using antiquated code
OSINT - Death Comes Calling: Thanatos/Alphabot Trojan Hits the Market
Threat Level 4
AlienVault | Cyber Attacks Leveraging the COVID-19/CoronaVirus Pandemic
AlienVault | Cybercriminals Prey on the COVID-19 Pandemic
AlienVault | Fake Corona Antivirus distributes BlackNET remote administration tool
AlienVault | Fake COVID-19 survey hides ransomware in Canadian university attack
Covid19 Stats Pakistan: Total 534, Deaths 3, Recovered 5
Covid19Malware feed
Cyber_Hat | APT29 targets COVID-19 vaccine development
Cyber_Hat | APT36 jumps on the coronavirus bandwagon, delivers CrimsonRAT
Cyber_Hat | attack activities using COVID
Grand Total

Fig. 2. Events with Threat Level

C. Situational Awareness with Smart Glasses

While smart glasses were initially used as simple front end displays, new devices have significantly better computing power and are able to perform more complex tasks. And as the technology is maturing, different types of AR devices and applications are becoming available to the mainstream. AR can already be effectively used to help in tasks like vehicle operating [9], [12], system monitoring [14], drone operating [3], military field operations [5] etc. The AR implementations most commonly used in different tasks involve projecting information or objects into the perceived real-world environment to assist and inform the user. More importantly AR can be used to help keep the user's focus in the important parts of the system. This is usually accomplished by gathering and projecting the system critical information, traditionally scattered in various places and/or multiple monitors, into the AR view. This approach frees the user from having to monitor multiple different places for the critical information, and they can more effectively focus on their main tasks.

AR implementations are often used as a tool to help the operator focus on a main task, rather than making AR itself the main focus. Pascale et al. [14] experimented using AR to keep nurses up to date on patients' vital signs in order to help them better focus on a main task of doing dosage calculations. This is similar to our experiment, where there is a main task that is done using a regular display, pen and paper, while monitoring other information via AR overlays. This kind of multitasking helper function spreads across many fields. For example, Rowen et al. [16] and Hong et al. [9] implemented AR in a similar fashion in order to help boat operators focus on their main task of navigation while Coleman et al. [3] used a similar method to help keep the focus of drone operators on the drone's video feed.

When it comes to Situational Awareness (SA), using an AR application can have a positive effect but it can also increase the attention demand as found in two studies on telemaintenance and teleoperation [1], [2]. In these experiments, a VR application actually performed better in term of SA and while the AR implementation improved SA it also increased attention demand more than the VR application. In many situations though, VR and AR both are not a feasible option. For vehicle operating and system monitoring AR is the far superior, and in most cases, the only realistic option. It has been shown that using AR for guidance, safety, and general information affects SA in a positive way in a maritime environment [9], [16] and driving a car on public roads [12]. Similarly, a positive SA effect was recorded in an experiment by [14], where a nurse was monitoring the vital information of several patients via AR while at the same time performing another task.

III. Methods

A. Participants

Given the COVID-19 limitations, we recruited a sample of convenience from the student population comprised of 6 participants in total (3 male and 3 female). The experiment was conducted at the Center for Ubiquitous Computing, University of Oulu, Finland.

B. Experimental Design and setup

In the experiment we tested an AR overlaid information interface coupled with a regular dual-monitor desktop setup. Testing was conducted using a within-subjects design with one condition, so each participant tested the same system and did the same tasks. The experiment simulates the work space of a cybersecurity analyst/monitor. The setup features a desktop computer with dual monitors, one of which is used for finding information and the other for experiment related recording etc. The AR device (HoloLens 2) overlays a threat level indicator in between the two monitors and an event information panel next to the left side monitor. This setup simulates multi tasking between a main working task using the keyboard, mouse, and the right side monitor and a secondary task of monitoring cybersecurity information presented on the left side panel. Participants were not able to make any adjustments to the location and height of either the virtual or the physical work station.

C. Experimental session

Before the explanation of the experiment, the participants signed a consent form and filled in their age and gender. The participants were told that they would assume the identity of a cybersecurity expert monitoring COVID-19-related cybersecurity information using HoloLens 2 AR smart glasses while at the same time completing a separate main task. The main task was to find answers to 10 pre-determined questions, using a provided web browser (right side monitor in Figure 3). The participants were also told that they could take notes about the constantly updating security events (for example write down event id numbers and corresponding threat level), but that it was up to them to decide whether to do this secondary task or not. Next, the AR implementation and the information it displayed was explained. The participants' desktop arrangement including the AR projected overlays can be seen in Figure 3. Lastly, the participants were informed that after the experiment, there would be a short questionnaire.

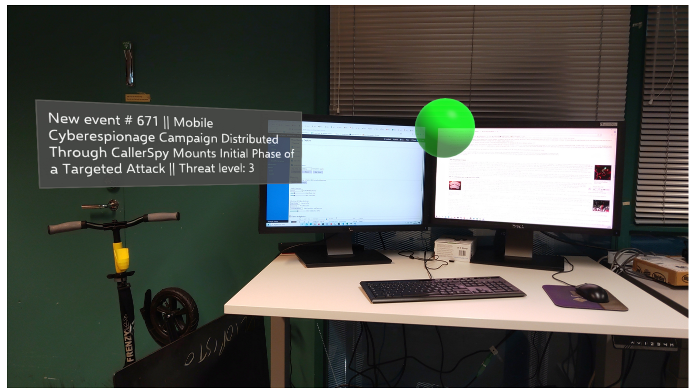


Fig. 3. Experiment setup with the threat level indicator in the middle (green sphere) and the event information panel on the left

After the experiment was started, the 10 questions were revealed to the participants. They were also provided a sheet of paper to use to answer the questions which also included a table to write down event ids and their corresponding threat levels if they chose to do so. The conducting researcher would silently follow the experiment and not talk to the participant unless they asked a question relating to the experiment procedure or if they needed help with something (for example got stuck in searching the answer for the first question). The experiment would last roughly 7 minutes, the time varying a bit depending on the accuracy of the randomized event finding algorithm (see next section). After the test was finished, the participants filled a Situational Awareness Rating Technique (SART) questionnaire. Video recordings were taken from 4 out of 6 experiments for verification purposes. The videos were recorded via HoloLens 2's integrated camera so the recordings would accurately show where the participants' view was focused at any point during the experiment. Before and after each experiment, the HoloLens 2 was sanitized using Cleanbox, a machine that uses ultraviolet light to kill bacteria, made specifically for disinfecting masks, eye-wear, and headsets.

To evaluate the SA of user during the experiments the Situation Awareness Rating Technique (SART) [19], which is a post-trial subjective rating technique, was administered to all participants [17]. The SART score is calculated with the formulas

$$SA = U–(D–S)$$

and here specifically:

$$\sum(Q8+Q9)-(\sum(Q1+Q2+Q3)-\sum(Q4+Q5+Q6+Q7))$$

where U is the summed understanding, D the summed demand, and S is the summed supply (limit -14 and 46).

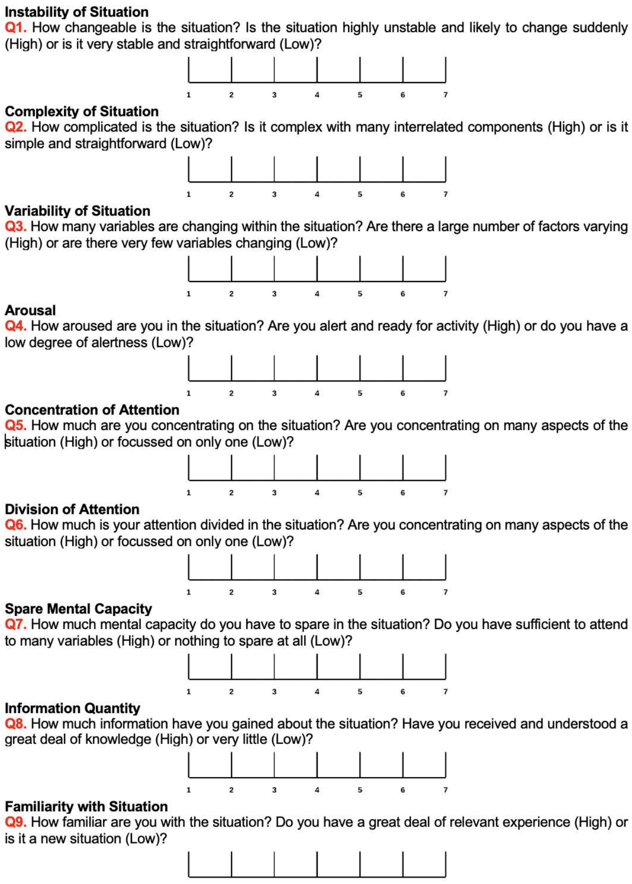


Fig. 4. SART post-simulation questionnaire

D. Development: Integration with the HoloLens 2

The HoloLens 2 application was created with Unity game engine using Mixed Reality Toolkit and C# scripts. For connecting to the COVID-19 database in MISP, UnityWebRequest was used. The MISP instance exposes an API which in return gives the caller a JSON-structured response. The data received via the UnityWebRequest is temporarily stored in the unit's memory. First, the data is parsed with SimpleJSON [7] to support exploration and iteration. Next, a custom algorithm is used to randomly pick 60 events, 15 for each of the 4 threat levels. The event information is then displayed on an AR projected panel situated roughly 45-degrees left of the user. Because the COVID-19 database consists mostly of level 4 threats, the system shows a maximum of 5 events, for each threat level, until moving onto a new cycle. This cycle is repeated 3 times for a total of 60 events (15 events * 4 threat levels). The event information panel displays each event for roughly 5 seconds. The threat level of the event is also visualized using a color-coded floating sphere that is projected in front of the user's viewpoint. The colors relate to the threat levels as follows: white = threat level 4; green = threat level 3; yellow = threat level 2; and red = threat level 1. Level 1 is the highest threat level and level 4 the lowest. The color of the ball and the event information are updated at the same time, so the user will know the threat level of the current event, just by glancing at the color of the object.

IV. Results

The gender differences were analysed using one-way analysis of variance (ANOVA) and the correlations between SART scores and task performance with Pearson's correlation test (*p*-value for both tests was chosen as 0.05). We are mostly interested in the SART scores as well as the performance in the 2 tasks. The final SART scores did not show any significant differences between the genders. But when looking at the individual components of the scores, some differences do stand out. More specifically there were significant differences between males and females in questions 5 ($p = 0.033$) and 6 ($p = 0.005$). For question 5: *"How much are you concentrating on the situation? Are you concentrating on many aspects of the situation or focused on only one?"* and question 6: *"How much is your attention divided in the situation? Are you concentrating on many aspects of the situation or focused on only one?"*, the female participants reported significantly lower values than their male counterparts (Table II). Lower values in this case meaning that the female participants thought their concentration was not as divided nor did they feel like they had to concentrate hard, and that they did not feel like they had to concentrate on many different aspects during the experiment.

TABLE I
Participant SART scores and task performance

#	Age	Gender	SART score	Answers	Events
1	27	M	11	7/10	14
2	29	M	16	7/10	13
3	29	F	5	2/10	0
4	26	F	7	3/10	0
5	31	M	11	4/10	4
6	27	F	14	1/10	13

When evaluating task performance, one category stands out as significant. The number of questions answered was significantly ($p = 0.026$) higher for the male group (Table II). And while the differences in SART scores and total tasks completed (answers + events) were not found to be significant ($p = 0.279$ and $p = 0.153$), they still indicate that there could be significant differences to be found if the identical experiment was carried out with a bigger sample.

The correlation between the SART score and task performance (for all participants) showed significant results in two relationships. The correlation between SART score + logged events (Pearson correlation = 0.852, $p = 0.031$) and SART score + total tasks (answers + events) (Pearson correlation = 0.822, p = 0.045) revealed a significant, yet weak, correlation between the results (Table III). These results indicate that the reported total SART score and task performance do in fact have a direct, albeit weak, correlation between them.

TABLE II
SIGNIFICANT DIFFERENCES BETWEEN GENDERS

Attribute	Mean for men	Mean for women	p
SART Q5	5.33	2.33	0.033
SART Q6	5.67	2.00	0.005
Number of answers	6	2	0.026

p value threshold = 0.05

Additionally when looking at the results, it can be seen that one participant in both gender groups performed differently to the other two. In the male group, one participant answered fewer questions and recorded fewer events than the other two, but still did so in equal amounts (4 answers and 4 events). In the female group, one concentrated singularly on the secondary event logging task and seemed to forget the main task of answering the questions. These findings are interesting because they do seem to indicate that in our experimental set up females only concentrated on a single task while males concentrated on both more equally.

TABLE III
SART SCORE CORRELATION WITH TASK PERFORMANCE METRICS

Relationship	Pearson correlation	p
SART score and number of answers	0.421	0.406
SART score and logged events	0.852	0.031
SART score and total tasks (answers + events)	0.822	0.045

p value threshold = 0.05

V. CONCLUSIONS AND FUTURE WORK

The main takeaway from the results is that there were some significant, albeit subtle, differences in how women an men performed in this experiment. It could be seen from the task performance that women concentrated on one task while men divided their attention between both of them. Interestingly, women still reported that they needed significantly less concentration and that their attention was significantly less divided than what the men said. The relationships between SART scores and two task performance metrics also showed significant correlation indicating that better situational awareness does translate to increased task performance. While there are other studies suggesting gender differences in aspects of task performance in multitasking situations, there is also very assuring results showing that there are no differences [8], which is an additional reason for remaining wary of the results we are seeing in our study.

Furthermore, the relatively small sample size does impose a limitation and thus, one should not draw any major conclusions from the results. Despite referring to the participants' gender in the above we do not propose there are conclusive gender differences based on our study. Nevertheless, because of the potential bias AR technology might introduce into workplace performance, the experiment should be repeated again with some improvements to the testing procedure, implementation and also with a larger sample size to verify the results.

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